Ballistocardiography sensor based on Polymeric Optical Fiber: A frequencies validation study

Juan Andrés García[a], Vinicius Mageste Coimbra[a], Daniel E. Garcia A.[a], Marcelo Eduardo Vieira Segatto[a], Carlos A. Cifuentes[b], and Camilo A. R. Diaz*[a]
[a]Telecommunications Laboratory (LabTel), Electrical Engineering Department, Federal University of Espírito Santo, Vitoria, Brazil.
[b]Bristol Robotics Laboratory, University of the West of England. United Kingdom

ABSTRACT

Wearable monitoring systems have represented a breakthrough in monitoring vital signs such as respiratory and heart rates. These systems can collect information from the vibrations in the chest due to blood pumping and the movements generated by breathing, known as ballistocardiography (BCG). This work presents the development of a sensor based on fiber optics, developing a sensitive area with a viscoelastic material. The sensor validation was performed with an experimental setup in which heart and breathing movements were simulated. Three combinations of respiratory and cardiac frequencies were selected to evaluate the sensor performance. The sensor's performance achieved an average success rate of 97% for pulse and 100% for respiration detection. It also demonstrated the ability to detect signal peaks at different magnitudes and frequencies. This makes the sensor attractive for further applications such as stress detection and evaluation of the response to several parameters to improve its performance.

Keywords: Heart rate, Optical fiber sensors, POF sensor, Respiration rate.

1. INTRODUCTION

Wearable systems for monitoring physiological parameters have a great capacity to provide important information regarding the health of users [1]. Electrocardiograms (ECGs) and photoplethysmography (PPG) are among the most used measurements for this type of device, usually integrated with chest or wrist devices to acquire the respective signals [2]. Besides these, ballistocardiography (BCG) has been used as an alternative for measuring both cardiac and respiratory parameters [3]. It measures the vibrations of the chest due to the pumping of the heart blood and the movement generated by respiration, which makes it suitable for estimating values such as heart rate (HR) and respiratory rate (RR) [4].

BCG signals can be captured with fiber optic sensors (OFS), which are characterized by their sensitivity, high stability, compact size, and electromagnetic immunity. Different operating principles can be used to measure physical, chemical, or environmental conditions, such as interferometry, fiber Bragg grating (FBG), specklegram, and intensity [5]. Among those, sensors based on intensity variations with polymer optical fiber (POF) have additional advantages such as low cost, ease of fabrication, simplicity to interrogate, and flexibility improving mechanical resistance [6].

POF-based sensors can be fabricated by creating sensitive areas where an intensity variation is generated due to mechanical deformations and/or temperature variations. The fabrication involves polishing the fiber's jacket and cladding, bending the fiber, and coupling parts, or bonding POFs through viscoelastic materials [7]. To measure the BCG signals, the sensor's sensitive area must be highly responsive to small disturbances caused by the intensity of pulse variations in the chest. BCG may have various health applications such as physiological monitoring, abnormal cardiac and respiratory pattern detection, and stress through physiological relationships with the autonomic nervous system [8].

In this context, this work presents the development of a low-cost sensor made of POF for cardiac and respiratory monitoring. The sensor was tested on a testbed under dynamic conditions to evaluate the sensor's ability to identify the frequencies and peaks of each signal i.e. cardiac and respiratory. Section 2 presents the POF sensor development and the experimental validation. Section 3 presents the result. Finally, the conclusion and future work are drawn in Section 4.

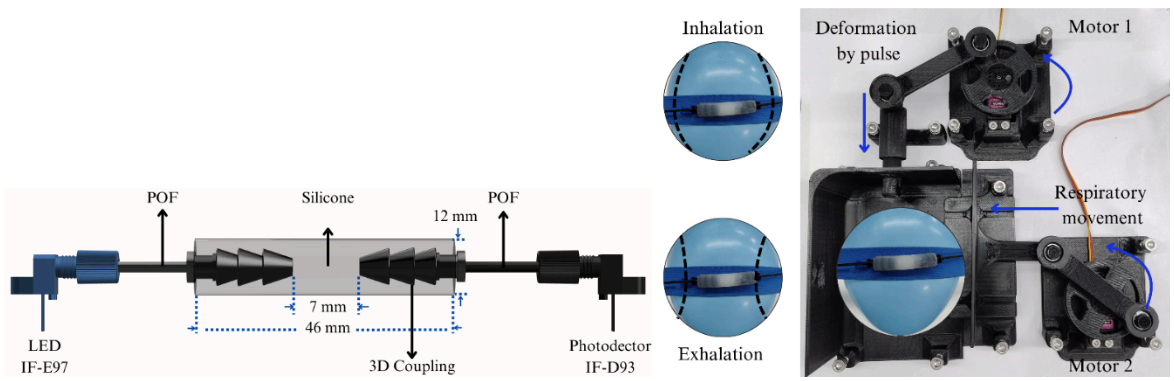
*camilo.diaz@ufes.br

29th International Conference on Optical Fiber Sensors, edited by José Luís Santos, Manuel Lopez-Amo Sainz, Tong Sun, Proc. of SPIE Vol. 13639, 136399C

2. METHODOLOGY

POF-based sensor: For sensor development (Figure 1a), a polymethyl-methacrylate (PMMA) POF (SH4001, Mitsubishi Chemical Co., Japan) was used. A light-emitting diode (LED, IF-E97, Industrial Fiber Optics, USA) was connected at one end of the POF, and the other end of the sensor was connected to a photodarlington (IF-D93, Industrial Fiber Optics, USA) to read the intensity variations due to perturbations in the sensitive area. This area was developed with EcoFlex silicone (Polisil, Brazil), a viscoelastic material that allows coupling the light from one POF to the other, keeps the POFs aligned, and increases the sensor's sensitivity due to its high capability to respond to deformations. In addition, to enhance the adhesion between silicone and POFs, small polylactic acid (PLA) supports were 3D printed and attached to the fiber ends, featuring protrusions for improved mechanical support.

For the shape of the silicone (a cylinder of 46 mm in length and 12 mm in diameter), a 3D printed mold was used, where a homogenized mixture of components A (silicone base) and B (catalyst) were joined in a 1A:1B ratio with contaminant-free elements. After one day of curing, the mold was removed to obtain the final shape of the sensor, as shown in Figure 1a. This silicone structure keeps the POFs aligned and depending on the deformation over this sensitive zone, intensity modulations are induced, i.e. by increasing the distance between the POFs, less optical power is coupled from the LED to the photodetector.



(a) (b)

Figure 1. a) Developed optical fiber sensor for pulse and breathing rate monitoring, b) Proposed setup for pulse and breathing frequencies simulation.

Experimental Setup: The experimental setup aims to simulate the body movements produced by breathing and those generated by cardiac activity to validate the BCG sensor. Two servomotors MG996 (TowerPro, China) were used, one in charge of simulating cardiac movements and the other one the respiratory movements. A mechanical mechanism allows to transform the motors' rotation into linear movements. To simulate the person's chest and the transmission of body movements to the sensor, a balloon was used due to its flexible behavior. The sensor was anchored to the balloon using an elastic Velcro, to which the movements generated by the motors were transmitted to the balloon (see Figure 1b). When monitoring these types of movements generated by cardiorespiratory activity, it is important to consider the amplitude and frequency of each of its components [9]. Therefore, for the simulation of respiratory movements a movable wall was used to press the balloon and for cardiac movements, a ram-shaped piece was used to generate disturbances in the balloon.

Procedure: For the performance sensor evaluation, three combinations of respiratory and cardiac frequencies were selected, and data was recorded for one minute three times for each combination. The selected frequencies are distributed as follows: test one with RR of 16 bpm and HR of 60 ppm; test two with RR of 18 bpm and HR of 86 ppm; and test three with RR of 20 bpm and HR of 100 ppm. These frequencies are also within the range of heart rate (0.5 Hz to 10 Hz) and respiratory rate (0.1 Hz to 0.3 Hz) [12]. The motors' control and the sensor data collection were performed with a microcontroller Teensy 3.6 (PJRC, Portland, OR, USA), which has an in-chip 16-bit resolution analog-to-digital converter. For signal processing, two order 2 Butterworth bandpass filters were used, with cutoff frequencies of 1 Hz to 3 Hz for pulse and 0.1 Hz to 0.8 Hz for respiration [7]. After this, a peak detection was performed for each signal and compared with the motor control signal. The processing was carried out in MATLAB R2024a (MathWorks, USA).

3. RESULTS

The sensor was subjected to a dynamic test to evaluate its deformation in load and unload. Figure 2a shows the hysteresis response of the sensor when subjected to strain. The maximum voltage variation generated by strain is 1.8 mV when deformations greater than 10 mε are applied. The hysteresis calculated was 0.003 V·mε. Figure 2b shows the signals used to control the motors in each test, which interact directly with the balloon. The sensor response is generated by the balloon interaction, where the balloon movements are related to the motor control signals. Figure 2c shows the signal collected from the sensor and filtered signals with their respective references. In the raw signal, there is a lower-frequency but higher-amplitude component (respiration), and high-frequency but lower-amplitude components (pulse). In the case of respiration, the behavior of the signal and the number of peaks detected are related to the input signal. In the case of the pulse signal, the peaks' amplitudes are not always the same; this is because the inhalation and exhalation movements generate changes in the shape of the balloon, which makes the ram-shaped piece not always hit with the same intensity. This is an important characteristic of the validation system, due to in a real scenario, the user will wear the sensor and will perform several activities, which can lead to non-homogeneous amplitude variations.

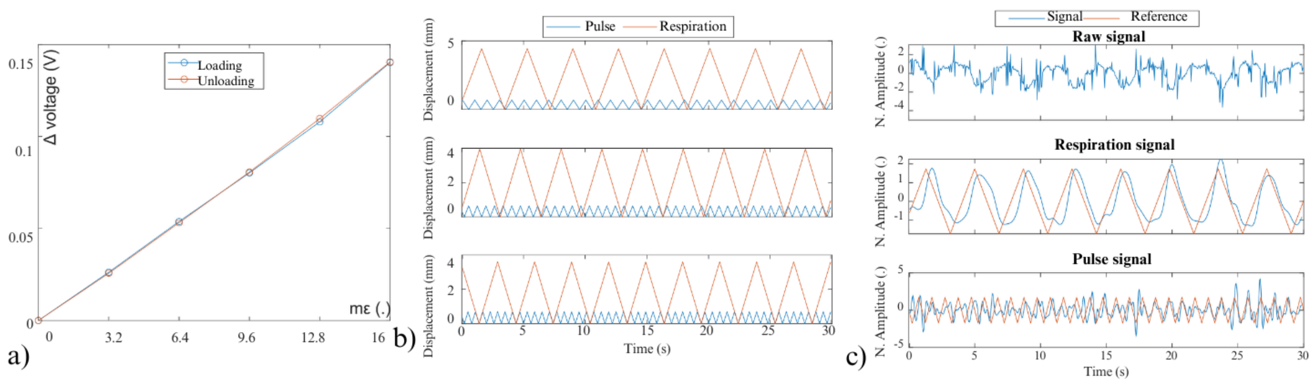


Figure 2. Sensor response and control signals. a) Hysteresis graph for the sensor, b) input signals for the three tests, c) sensor example response for 30 s.

Table 1 shows the results of all the tests. For respiratory frequencies, the sensor was able to detect the respiration rate at all frequencies correctly. In the case of the pulse, the values obtained were greater than 97%. It can be seen that the time difference for the pulse in all experiments is very similar, which shows a delay between the input signals and those read by the sensor independent of the frequencies used. This behavior can also be observed in the respiration signal, suggesting the delay is given by the mechanical system that transforms the rotational movements into linear movements. In general, the sensor can detect signals with different frequencies and amplitudes.

Table 1. Multiple frequencies response of the sensor.

	Frequency 1			**Frequency 2**			**Frequency 3**		
	Peaks		**Time (s)**	**Peaks**		**Time (s)**	**Peaks**		**Time (s)**
	Reference (BPM)	**Obtained (BPM)**		**Reference (BPM)**	**Obtained (BPM)**		**Reference (BPM)**	**Obtained (BPM)**	
Pulse	60	61±1.53	0.22±0.13	86	85±1.52	0.2±0.11	100	97±1.53	0.22±0.14
Respiration	16	16±0.57	0.29±0.11	18	18±0	0.32±0.07	20	20±1	0.32±0.42

4. DISCUSSION

By using viscoelastic materials such as EcoFlex for the development of sensors, it is possible to determine the peaks of respiration and pulse signals, providing information on both frequencies and time differences between peaks. This information is useful for measurements such as heart rate variability (HRV) for subsequent stress evaluations [7].

FBG-based sensors are broadly used in pulse detection applications due to their high sensitivity. However, it can be observed that the POF-based sensor can also measure this type of signal with high accuracy, even without the need for complex interrogation systems [10]. In the case of heart rate measurement, the lowest success rate obtained was 97%, a value that can be compared with that obtained with FBGs when testing users under laboratory conditions (95%) [11].
In the controlled testing of both ECG and PPG measurements, standard equipment is commonly used to perform verifications and assess performance before implementation in users or to ensure proper operation [13], [14]. However, this practice is not yet widespread for BCG sensors. Establishing setups to simulate and test such sensors is a foundational step for future applications, enabling the incorporation of more desired features and the simulation of various behaviors or patterns. Among the limitations of this study was the influence of environmental factors such as ambient light, which makes coating necessary for further applications. Another limitation with the measured time between the motor signal and the measured signal is the mechanical motion conversion system, which may have a delay between when the motor signal is output and when the wall pushes the balloon or the stick hits the balloon, so 0.2s - 0.3s of delay may be due to transfer to the sensor.

5. CONCLUSION AND FUTURE WORK

In this work, an OFS was developed to monitor cardiac and respiratory activities simultaneously. The proposed sensor was able to obtain the information corresponding to the respiratory and cardiac frequencies in a testbed, in addition to correctly identifying the time between the peaks. The relative error for 60, 86, and 100 bpm for pulse were 1.67%, 1.16%, and 3%, respectively. For respiration (16, 18, 20 bmp) there were no errors. For the three tests, the time differences are very similar for both, respiration with 0.32 s and pulse with 0.22 s. The proposed testbed allowed the validation of the sensor under dynamic conditions by applying different frequencies and amplitudes of each signal, where the sensor was able to detect correctly both signals. Future work will focus on studying the implications of geometry, viscoelastic materials, light source, fiber spacing, and other parameters to improve sensor performance in terms of sensitivity and accuracy. In addition to making implementations in real scenarios with users under several activities and further evaluation of physiological stress.

ACKNOWLEDGMENTS

This work is supported by FAPES (EXTRATO ATA DA 6ª REUNIÃO CCAF/2022 – "CPID 2.2"). Camilo A. R. Diaz acknowledges the financial support of FAPES (459/2021, 2022-97F12), CNPq (310668/2021-2, 404111/2023-8) and IEEE RAS SPARX, FINEP (Tec assistiva − 2784/20, Tec assistiva 2 − 2132/22).

REFERENCES

[1] Hinde, K., White, G., & Armstrong, N. (2021). Wearable devices suitable for monitoring twenty-four-hour heart rate variability in military populations. *Sensors (Basel)*, *21*(4), 1061.
[2] Kim, K. B., & Baek, H. J. (2023). Photoplethysmography in Wearable Devices: A Comprehensive Review of Technological Advances, Current Challenges, and Future Directions. Electronics, 12(13), 2923.
[3] Gardner, M., Randhawa, S., Malouf, G., & Reynolds, K. (2021). A wearable ballistocardiography device for estimating heart rate during positive airway pressure therapy: Investigational study among the general population. *JMIR Cardio*, *5*(1), e26259.
[4] Han, X., Wu, X., Wang, J., Li, H., Cao, K., Cao, H., Zhong, K., & Yang, X. (2021). The Latest Progress and Development Trend in the Research of Ballistocardiography (BCG) and Seismocardiogram (SCG) in the Field of Health Care. Applied Sciences, 11(19), 8896.
[5] He, R., Shen, L., Wang, Z., Wang, G., Qu, H., Hu, X., & Min, R. (2023). Optical fiber sensors for heart rate monitoring: A review of mechanisms and applications. *Results in Optics*, *11*, 100386.
[6] Leal-Junior, A. G., Díaz, C. R., Leitão, C., Pontes, M. J., Marques, C., & Frizera, A. (2019). Polymer optical fiber-based sensor for simultaneous measurement of breath and heart rate under dynamic movements. *Optics & Laser Technology*, *109*, 429–436.
[7] Gaitán-Padilla, M., Múnera, M., Pontes, M. J., Vieira Segatto, M. E., Cifuentes, C. A., & Díaz, C. A. R. (2024). Development of a polymeric optical fiber sensor for stress estimation: A comparative analysis between physiological sensors. *IEEE Sensors Journal*, *24*(20), 32140–32149.
[8] Dalmeida, K. M., & Masala, G. L. (2021). HRV Features as Viable Physiological Markers for Stress Detection Using Wearable Devices. Sensors, 21(8), 2873.
[9] Centracchio, J., et al. (2022). Respiratory-induced amplitude modulation of forcecardiography signals. *Bioengineering (Basel)*, *9*(9), 444.
[10] Lo Presti, D., Massaroni, C., D'Abbraccio, J., Saccomandi, P., Formica, D., & Schena, E. (2019). Wearable system based on flexible FBG for respiratory and cardiac monitoring. *IEEE Sensors Journal*, *19*(17), 7391–7398.
[11] Fajkus, M., Massaroni, C., D'Abbraccio, J., Saccomandi, P., Formica, D., & Schena, E. (2024). FBG sensor for heart rate monitoring using 3D printing technology. *IEEE Access*, *12*, 57150–57162.
[12] Tavares, C., Leitão, C., Lo Presti, D., Domingues, M. F., Alberto, N., Silva, H., & Antunes, P. (2022). Respiratory and heart rate monitoring using an FBG 3D-printed wearable system. *Biomed. Opt. Express*, *13*, 2299–2311.
[13] Quiroz-Juárez, M. A., Rosales-Juárez, J. A., Jiménez-Ramírez, O., Vázquez-Medina, R., & Aragón, J. L. (2022). ECG Patient Simulator Based on Mathematical Models. Sensors, 22(15), 5714.
[14] Pittella, E., Testa, O., Podestà, L., & Piuzzi, E. (2024). An Optical Signal Simulator for the Characterization of Photoplethysmographic Devices. Sensors, 24(3), 1008.